



Matrix Theory

15 Least Squares

Dr. Konda Reddy Mopuri
Dept. of AI, IIT Hyderabad
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Until now



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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- $Ax = b$ either has a solutions or not

Until now



- $Ax = b$ either has a solutions or not
- If b is not in $C(A)$ no solution (inconsistent; Elimination fails)

Until now



- $Ax = b$ either has a solutions or not
- If b is not in $C(A)$ no solution (inconsistent; Elimination fails)
- Inconsistency is more likely when there are more equations than variables

$$m > n$$

$$\begin{bmatrix} \end{bmatrix}$$

A

Example



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$$2x = b_1$$

$$3x = b_2$$

$$4x = b_3$$

$$b_1 : b_2 : b_3 = 2 : 3 : 4$$

Example



$$b = c \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

$$2x = b_1$$

$$3x = b_2$$

$$4x = b_3$$

- Solution exists if b lies on the same column as $a = (2, 3, 4)$



Inconsistency is practical



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- Despite their unsolvability, they exist in practice

Inconsistency is practical



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- Despite their unsolvability, they exist in practice
- What to do?

Inconsistency is practical



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- Despite their unsolvability, they exist in practice
- What to do?
- Can we consider a subset of the system of equations? → no error on some, large error on others

$$Ax = b \rightarrow$$

↑

Inconsistency is practical



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!

- Hard to justify when all the equations come from the same system

Inconsistency is practical



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- Hard to justify when all the equations come from the same system
- Choose the x that minimizes “an average error” in all equations

Average Error



- Most convenient form of an average error:

$$E^2 = (2x - b_1)^2 + (3x - b_2)^2 + (3x - b_3)^2$$



Average Error



- Most convenient form of an average error:
$$E^2 = (2x - b_1)^2 + (3x - b_2)^2 + (3x - b_3)^2$$
- Called, the sum of squares error (SSE)

MSE

RSE

Sum of squared
errors

Average Error



- Most convenient form of an average error:

$$E^2 = (2x - b_1)^2 + (3x - b_2)^2 + (3x - b_3)^2$$

- Called, the sum of squares error (SSE) ✓
- Its minimum value is 0 $\rightarrow$ exact solution ✓

$b \in C(A)$

≥ 0 $b \notin C(A)$

Average Error



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- In case of b not proportional to a , error E^2 is a parabola

Average Error



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- In case of b not proportional to a , error E^2 is a parabola
- Where does the minimum error occur?

Average Error



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- In case of b not proportional to a , error E^2 is a parabola
- Where does the minimum error occur?
- At the lowest point, where the derivative is zero

Average Error



$$E^2 = (2x - b_1)^2 + (3x - b_2)^2 + (3x - b_3)^2$$

$$\frac{dE^2}{dx} = 2[(2x - b_1)2 + (3x - b_2)3 + (3x - b_3)4] = 0$$

LS solution



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$$|2x - b_1| + |3x - b_2| + |4x - b_3|$$

$$E^2 = (2x - b_1)^2 + (3x - b_2)^2 + (3x - b_3)^2$$

$$\frac{dE^2}{dx} = 2[(2x - b_1)2 + (3x - b_2)3 + (4x - b_3)4] = 0$$

LS solution of this system $ax = b$, denoted by $\hat{x} = \frac{2b_1 + 3b_2 + 4b_3}{2^2 + 3^2 + 4^2} = \frac{a^T b}{a^T a}$

General case



- Solve $ax = b$ by minimizing $\|ax - b\|^2$

General case



- Solve $ax = b$ by minimizing $\|ax - b\|^2$
- $E^2 = \sum_i^m (a_i x - b_i)^2 = (a_1 x - b_1)^2 + \dots + (a_m x - b_m)^2$

General case



- Solve $ax = b$ by minimizing $\|ax - b\|^2$
- $E^2 = \sum_i^m (a_i x - b_i)^2 = (a_1 x - b_1)^2 + \dots + (a_m x - b_m)^2$
- The derivative of E^2 is zero at the point $\hat{x}$, if
 $(a_1 \hat{x} - b_1)a_1 + (a_2 \hat{x} - b_2)a_2 + \dots + (a_m \hat{x} - b_m)a_m = 0$

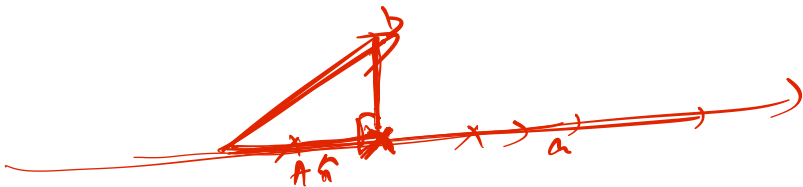


General case



- Solve $ax = b$ by minimizing $\|ax - b\|^2$
- $E^2 = \sum_i^m (a_i x - b_i)^2 = (a_1 x - b_1)^2 + \dots + (a_m x - b_m)^2$
- The derivative of E^2 is zero at the point $\hat{x}$, if
$$(a_1 \hat{x} - b_1)a_1 + (a_2 \hat{x} - b_2)a_2 + \dots + (a_m \hat{x} - b_m)a_m = 0$$
- We are minimizing the distance from b to the line through a !

$$\hat{x} = \frac{a^T b}{a^T a}$$



$$\underline{ax = b}$$

General case



- We are minimizing the distance from b to the line through a
- LS solution to $ax = b$ in one unknown is

$$\hat{x} = \frac{a_1 b_1 + a_2 b_2 + \dots + a_m b_m}{a_1^2 + a_2^2 + \dots + a_m^2} = \frac{a^T b}{a^T a}$$





Least Squares problem with multiple variables

LS with multiple variables



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- Instead of one column and one variable, we have $Ax = b$, where $A_{m \times n}$ and $(m > n)$

LS with multiple variables



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- Instead of one column and one variable, we have $Ax = b$, where $A_{m \times n}$ and $(m > n)$
- It is expected to be inconsistent (b is outside $C(A)$)

LS with multiple variables



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- Problem is to find $\hat{x}$ that minimizes the error



LS with multiple variables



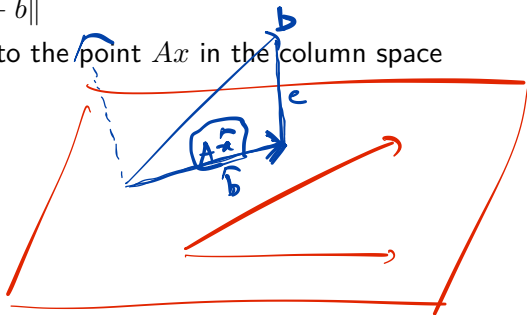
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- Problem is to find $\hat{x}$ that minimizes the error
- Error $E = \|Ax - b\|$

LS with multiple variables



- Problem is to find $\hat{x}$ that minimizes the error
- Error $E = \|Ax - b\|$
- Distance from b to the point $A\hat{x}$ in the column space



LS with multiple variables



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Searching for the least-squares solution $\hat{x}$, which minimizes E , is the same as locating the point $p = A\hat{x}$ that is closer to b than any other point in the column space.

LS with multiple variables



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- p must be the projection of b on to the column space of A

LS with multiple variables



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- p must be the projection of b on to the column space of A
- The error $b - A\hat{x}$ is orthogonal to the space.

LS with multiple variables

- p must be the projection of b on to the column space of A
- The error $b - A\hat{x}$ is orthogonal to the space. ✓
- It must be orthogonal to each column $a_1, a_2, \dots, a_n$ of A

$$\begin{array}{c} \underline{a_1^T(b - A\hat{x})} = 0 \\ \vdots \\ \underline{a_n^T(b - A\hat{x})} = 0 \end{array} \quad \text{or} \quad \underbrace{\begin{bmatrix} a_1^T \\ \vdots \\ a_n^T \end{bmatrix}} \begin{bmatrix} b - A\hat{x} \end{bmatrix} = 0. \quad \checkmark$$

LS with multiple variables



- p must be the projection of b on to the column space of A
- The error $b - A\hat{x}$ is orthogonal to the space.
- It must be orthogonal to each column $a_1, a_2, \dots, a_n$ of A

$$\begin{array}{ccc} a_1^T(b - A\hat{x}) = 0 & & \\ \vdots & \text{or} & \begin{bmatrix} a_1^T \\ \vdots \\ a_n^T \end{bmatrix} \begin{bmatrix} b - A\hat{x} \end{bmatrix} = 0 \\ a_n^T(b - A\hat{x}) = 0 & & \end{array}$$

- $A^T(b - A\hat{x}) = \mathbf{0}$ and $A^T A\hat{x} = A^T b$

$Ax = b$
 $\downarrow$
 x

LS with multiple variables



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$$E = \|Ax - b\|^2 \\ = (Ax - b)^T (Ax - b)$$

- Calculus way: partial derivative of $E^2 = (A\hat{x} - b)^T (A\hat{x} - b)$

LS with multiple variables



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- Calculus way: partial derivative of $E^2 = (A\hat{x} - b)^T(A\hat{x} - b)$ ✓
- This gives, $2A^T A\hat{x} - 2A^T b = 0 \rightarrow \underline{A^T A\hat{x} = A^T b}$ *minima*

Normal equations



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- $A^T A \hat{x} = A^T b$



Normal equations



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- $A^T A \hat{x} = A^T b$
- Solution has the property that the residual vector $e = b - A\hat{x}$ is orthogonal/normal to the column space of A

Normal equations



- $A^T A \hat{x} = A^T b$
- Solution has the property that the residual vector $e = b - A\hat{x}$ is orthogonal/normal to the column space of A
- It's **normal** to the plane spanned by the predictors.

LS with multiple variables



When $Ax = b$ is inconsistent, its least-squares solution minimizes $\|Ax - b\|^2$.

Normal equations: $A^T A \hat{x} = A^T b$

LS with multiple variables



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$$\underline{A^T A \hat{x} = A^T b}$$

$$A^T A_{n \times n}$$

$$\leftarrow A_{m \times n}$$

When $Ax = b$ is inconsistent, its least-squares solution minimizes $\|Ax - b\|^2$:

Normal equations: $A^T A \hat{x} = A^T b$

$A^T A$ is invertible if A is full-column rank. The best estimate $\hat{x} = \underline{(A^T A)^{-1} A^T b}$

$$\text{If } A^T A = 0 \quad \text{or} \quad A^T A x = 0 \quad \Rightarrow \quad \|Ax\|^2 = 0 \quad \Rightarrow \quad Ax = 0$$

$$\text{If } Ax = 0 \quad \text{then} \quad A^T A x = 0$$

LS with multiple variables



When $Ax = b$ is inconsistent, its least-squares solution minimizes $\|Ax - b\|^2$:

Normal equations: $A^T A \hat{x} = A^T b$

$A^T A$ is invertible if A is full-column rank. The best estimate $\hat{x} = (A^T A)^{-1} A^T b$

Projection of b onto the column space is the nearest point $A\hat{x}$:

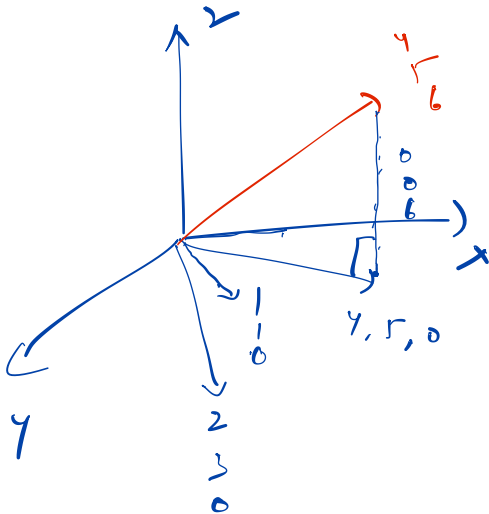
$$p = A\hat{x} = A(A^T A)^{-1} A^T b$$

Example



$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

- $Ax = b$ has no solution



Example



$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

- $Ax = b$ has no solution
- $A^T A \hat{x} = A^T b$ gives the best solution ✓

Example



$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

- $Ax = b$ has no solution
- $\underline{A^T Ax} = \underline{A^T b}$ gives the best solution
- $C(A)$ is the xy plane in $3D$ ✓

Example



$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

- $Ax = b$ has no solution
- $A^T Ax = A^T b$ gives the best solution
- $C(A)$ is the xy plane in $3D$
- Projection of $b = (4, 5, 6)$ onto this is $(4, 5, 0)$ ✓

Example

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$



- Let's confirm by working with normal equations

Example

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$



$$\underline{A^T A \hat{x} = A^T b}$$

- Let's confirm by working with normal equations

- $A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}$ ✓

Example



$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

- Let's confirm by working with normal equations

- $A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}$

- $A^T b = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 9 \\ 23 \end{bmatrix}$

Example



$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

- Let's confirm by working with normal equations

- $A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}$

- $A^T b = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 9 \\ 23 \end{bmatrix}$

- Solve for $\hat{x}$ $\begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix} \begin{vmatrix} 9 \\ 23 \end{vmatrix}$

$$R_2 = -4.5 R_1 + R_2$$
$$x_1 = -2 \quad x_2 = 1$$

Example



$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

- Let's confirm by working with normal equations

- $A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ 5 & 13 \end{bmatrix}$

- $A^T b = \begin{bmatrix} 1 & 1 & 0 \\ 2 & 3 & 0 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 9 \\ 23 \end{bmatrix}$

- Solve for $\hat{x}$ $\begin{bmatrix} 2 & 5 & | 9 \\ 5 & 13 & | 23 \end{bmatrix}$

- Projection $p = A\hat{x}$

$$\begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \\ 0 \end{bmatrix}$$

Example



$$A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 0 & 0 \end{bmatrix} \quad b = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

- Does the third equation matter?

$$\begin{array}{r} ||x+0y-6||^2 \quad E_3 \\ \hline ||x+2y-4||^2 \quad E_1 \\ + \\ ||x+3y||^2 \quad E_2 \\ -5 \\ = SSE \end{array}$$

What if b lies in $C(A)$

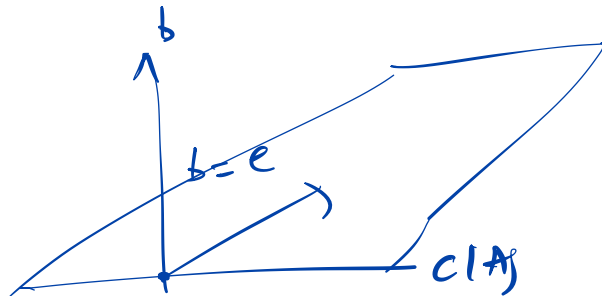


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What if b is orthogonal to the columns of A



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What if A is square and invertible?



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$$A^T A \hat{x} = A^T b$$

$$\begin{aligned}\hat{x} &= \frac{(A^T A)^{-1} A^T b}{=} \bar{A}^{-1} \bar{A}^T A^T b \\ &= \bar{A}^{-1} b\end{aligned}$$

What if A is a column?



$$\underline{A^T A \hat{x} = A^T b}$$

$$n=1$$

$$\hat{x} = \frac{A^T b}{A^T A}$$

$$= \frac{a^T b}{a^T a}$$

Cross-Product Matrix $A^T A$



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$n \times m$ $m \times n$

- Symmetric

(i,j) entry

Cross-Product Matrix $A^T A$



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- Symmetric ✓
- $A^T A$ has the same nullspace as A (nullspaces are identical) ✓

Cross-Product Matrix $A^T A$



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- Symmetric
- $A^T A$ has the same nullspace as A (nullspaces are identical)
- $\rightarrow$ if the columns of A are independent, then $A^T A$ is invertible ✓

Projection Matrix P



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- Closest point to b is $p = \underline{A(A^T A)^{-1} A^T b}$

Projection Matrix P



P

- Closest point to b is $p = A(A^T A)^{-1} A^T b$.
- The projection matrix $P = A(A^T A)^{-1} A^T$

Projection Matrix P



- Closest point to b is $p = A(A^T A)^{-1} A^T b$
- The projection matrix $P = \underline{A(A^T A)^{-1} A^T}$
- Projects any vector b onto the $C(A)$



Projection Matrix P



- Closest point to b is $p = A(A^T A)^{-1} A^T b$
- The projection matrix $\underline{P} = A(A^T A)^{-1} A^T$
- Projects any vector b onto the $C(A)$
- $\underline{e} = \underline{b} - \underline{P}b$ is the component in the orthogonal complement

$$\underline{(\underline{I} - \underline{P})}$$

Projection Matrix P



- Closest point to b is $p = A(A^T A)^{-1} A^T b$
- The projection matrix $P = A(A^T A)^{-1} A^T$
- Projects any vector b onto the $C(A)$
- $e = b - Pb$ is the component in the orthogonal complement
- $P^2 = P$ and $P^T = P$

Next



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- LS fitting of data

Next



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- LS fitting of data
- Orthogonalization (Gram-Schmidt)

Rough Work



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